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Batch: 15

Experiment-45

Algorithm

Main Program

1. Start.
2. Declare an integer variable `choice`.
3. Display the menu:
    - 1. Change date format
    - 2. Print all substrings
    - 3. Exit
4. Read the user's choice.
5. Use a `switch` statement:
    - If `choice = 1`, call `changeDateFormat()`.
    - If `choice = 2`, call `printSubstrings()`.
    - If `choice = 3`, exit the program.
    - Otherwise, display **"Invalid choice! Please try again."**
6. Repeat the menu until the user selects `3`.
7. Stop.

Function 1: `changeDateFormat()`

1. Start the function.
2. Declare `day`, `month`, and `year`.
3. Store the month abbreviations `Jan, Feb, Mar, ..., Dec` in a string array.
4. Read the date in `dd/mm/yyyy` format.
5. Check whether the month is between `1` and `12`.
6. If valid:
    - Use `month - 1` to select the corresponding month abbreviation.
    - Use `year % 100` to obtain the last two digits of the year.
    - Display the date in `dd-mmm-yy` format.
7. Otherwise, display **"Invalid month!"**
8. Return to the main program.

Function 2: `printSubstrings()`

1. Start the function.
2. Read the input string.
3. Find the length of the string using `strlen()`.
4. Use an outer loop `i` from `0` to `n-1` to select the starting position.
5. Use a second loop `j` from `i` to `n-1` to select the ending position.
6. Use a third loop `k` from `i` to `j` to print the characters of the substring.
7. Print a new line after each substring.
8. Repeat until all possible contiguous substrings are printed.
9. Return to the main program.

Test Case Table

Test Case	Menu Choice	Input	Expected Output
1	1	`12/10/2026`	`12-Oct-26`
2	1	`25/12/2025`	`25-Dec-25`
3	1	`01/01/2027`	`01-Jan-27`
4	1	`15/13/2026`	`Invalid month!`
5	2	`abc`	`a, ab, abc, b, bc, c`
6	2	`ab`	`a, ab, b`
7	2	`hello`	`h, he, hel, hell, hello`
8	3	—	`Exiting program...`
9	5	—	`Invalid choice! Please try`

C Program

```
#include <stdio.h>

#include <string.h>

void changeDateFormat()

{

   int day, month, year;

   char months[12][4] = {

      "Jan", "Feb", "Mar", "Apr",

      "May", "Jun", "Jul", "Aug",

      "Sep", "Oct", "Nov", "Dec"
```

```
    };

    printf("Enter date in dd/mm/yyyy format: ");
    scanf("%d/%d/%d", &day, &month, &year);

    if (month >= 1 && month <= 12)
    {
        printf("Converted date: %02d-%s-%02d\n",
                day, months[month - 1], year % 100);
    }
    else
    {
        printf("Invalid month!\n");
    }
}

void printSubstrings()
{
    char str[100];
    int i, j, k;
    int n;

    printf("Enter a string: ");
    scanf("%99s", str);
```

```
    n = strlen(str);

    printf("All substrings are:\n");

    for (i = 0; i < n; i++)
    {
        for (j = i; j < n; j++)
        {
            for (k = i; k <= j; k++)
            {
                printf("%c", str[k]);
            }
            printf("\n");
        }
    }
}

int main()
{
    int choice;

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Change date format\n");
```

```
    printf("2. Print all substrings of a string\n");
    printf("3. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            changeDateFormat();
            break;

        case 2:
            printSubstrings();
            break;

        case 3:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
    }

} while (choice != 3);
```

```
    return 0;

}
```

```
┌──(kali㉿kali)-[~]
└─$ vi hdoc45.c

┌──(kali㉿kali)-[~]
└─$ gcc hdoc45.c -o hdoc45

┌──(kali㉿kali)-[~]
└─$ ./hdoc45

──── MENU ────
1. Change date format
2. Print all substrings of a string
3. Exit
Enter your choice: 1
Enter date in dd/mm/yyyy format: 13/08/2008
Converted date: 13-Aug-08

──── MENU ────
1. Change date format
2. Print all substrings of a string
3. Exit
Enter your choice: 2
Enter a string: hello
All substrings are:
h
he
hel
hell
hello
```

```
e
el
ell
ello
l
ll
llo
l
lo
o

──── MENU ────
1. Change date format
2. Print all substrings of a string
3. Exit
Enter your choice: 3
Exiting program...
```